Day 26 - 8.1.1 Polynomial Investigation

Friday, April 21, 2023 12:34 PM

Objective: • Students will be able to … ○ Investigate properties of Polynomial equations ○ Determine what makes a graph bounce, wiggle, or cross the x-axis ○ Determine the degree of a polynomial ○ State the Fundamental Theorem of Algebra **Mathematical Practice(s):** reason abstractly and quantitatively, attend to precision, look for and make use of structure **Classwork:** • Bellringer: Check for Understanding : Polynomials • Polynomial Investigation • Whole Class Debrief • Vocabulary Notes • Exit Ticket **Homework:** • Day 26 Homework - Growth Check - Due Next Class **Announcement:** • Day 29: Trigonometry KC #2 ODD - Wednesday, April 15th EVEN - Thursday, April 16th	**Key Questions:** 1. What can we predict from looking at the equation of a polynomial? Why does this make sense? 2. Which form of a polynomial equation is most useful for making a graph (factored form or standard form)? What information does it give? 3. How does changing the exponent on one of the **factors** change the graph? a) Write a summary statement about what you noticed. Your conjecture may begin with: *For a function of the form $f(x) = (x - a)^n$ where n is a positive **even** integer…* b) Write a summary statement about what you noticed. Your conjecture may begin with: *For a function of the form $f(x) = (x - a)^n$ where n is a positive **odd** integer…* 4. How can we find the y-intercepts of the polynomial?

Day 26 Notes

Algebra 2 Honors Name: _____ P:____

Day 26 | Polynomial Investigation

Warm-Up: In your teams, read the following information.

METHODS AND MEANINGS

MATH NOTES

Vocabulary for Expressions

A mathematical expression is a combination of numbers, variables, and operation symbols. Addition and subtraction separate expressions into parts called terms. For example, $4x^2 - 3x + 6$ is an expression. It has three terms: $4x^2$, $3x$, and 6. The **coefficients** of the terms with variables are 4 and -3. 6 is called a **constant term**.

A single-variable **polynomial** is an expression that involves, at most, the operations of addition, subtraction, and multiplication. Most of the polynomials you will work with can be written as expressions with terms of the following form:

$$(\text{any real number})x^{(\text{whole number})}$$

For example, $4x^2 - 3x^1 + 6x^0$ is a polynomial, as is the simplified form, $4x^2 - 3x + 6$. Also, since $6x^0 = 6$, 6 itself is a polynomial.

The function $f(x) = 7x^5 + 2.5x^3 - x + 7$ is a polynomial function.

A **binomial** is a polynomial with only two terms, for example, $x^3 - 0.5x$ and $2x + 5$.

The following expressions are *not* polynomials: $2^x - 3$, $\frac{1}{x^2-2}$, and $\sqrt{x-2}$.

An expression that can be written as the quotient of two polynomials is a **rational expression**. For example, $\frac{1}{x^2-2}$ is a rational expression.

Which of the following equations are polynomial functions? For each one that is not, justify why not.

(a) $f(x) = 8x^5 + x^2 + 6.5x^4 + 6$

(b) $y = \frac{3}{5}x^6 + 19x^2$

(c) $y = 2^x + 8$

(d) $f(x) = 9 + \sqrt{x} - 3$

(e) $P(x) = 7(x - 3)(x + 5)^2$

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METHODS AND MEANINGS

MATH NOTES

Polynomials, Degree, Coefficients

Refer to the Math Notes box in Lesson 3.1.3 for an explanation of a **polynomial** in one variable.

Polynomials with one variable (often x) are usually arranged with powers of x in order, starting with the highest, left to right.

The highest power of the variable in a polynomial of one variable is called the **degree** of the polynomial. The numbers that multiply each term are called **coefficients**. See the examples below.

Example 1: $f(x) = 7x^5 + 2.5x^3 - \frac{1}{2}x + 7$ is a polynomial function of degree 5 with coefficients $7, 0, 2.5, 0, -\frac{1}{2}$, and 7. Note that the last term, 7, is called the **constant term** but represents the variable expression $7x^0$, since $x^0 = 1$.

Example 2: $y = 2(x + 2)(x + 5)$ is a polynomial in factored form with degree 2 because it can be written in standard form as $y = 2x^2 + 14x + 20$. It has coefficients 2, 14, and 20.

Create a polynomial for each of the following descriptions:

(a) Polynomial of degree 7 with leading coefficient -2 with 3 terms

(b) Polynomial with 2 terms and the largest degree of 3, leading coefficient of 4 and a constant of π

(c) Polynomial of degree 4, leading coefficient of 2

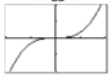
(d) Polynomial of degree 6, constant of 7, and leading coefficient of -3

Example:

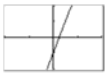
Equation	Graph	Zero A	Zero B	Zero C	Left End Behavior	Right End Behavior	Sign of Leading Coefficient	Degree
1] $y = 2x^3(x-1)(x+1)^2$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				

Bounce

Wiggle



Cross



Solve the equation to find the zeros by using the zero-product property. Then use Desmos to sketch the graph. Change the window settings to show -2.5 to 2.5 on each axis.

Each graph will have one or more of these zeros. If it has that zero, does it wiggle, bounce, or cross at that point? Give the exponent of the corresponding factor.

Describe the characteristics of the polynomial below. Are there multiple ways to find the degree of the polynomial?

Equation	Graph	Zero A	Zero B	Zero C	Left End Behavior	Right End Behavior	Sign of Leading Coefficient	Degree
1] $y = 2x^4(x-1)$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
2] $y = -6x(x+1)^5$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
3] $y = 3x(x-1)^3(x+1)^2$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				

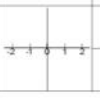
Equation	Graph	Zero A	Zero B	Zero C	Left End Behavior	Right End Behavior	Sign of Leading Coefficient	Degree
4] $y = -5x(x-1)^6(x+1)^7$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
5] $y = -5x^2(x+1)^3(x-1)^4$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
6] $y = 4x^2(x-1)^5(x+1)$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
7] $y = 200x^2(x-1)^3(x+1)^0$		bounce	bounce	bounce				

6] $y = 4x^2(x-1)^5(x+1)$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				
7] $y = 200x^2(x-1)^5(x+1)^6$		bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:	bounce wiggle cross none Exponent:				

8] What do you notice about the exponent on the factor in each equation when the graph **"crosses"** through a zero?	9] What do you notice about the exponent on the factor in each equation when the graph **"bounces"** off a zero?	10] What do you notice about the exponent on the factor in each equation when the graph **"wiggles"** through a zero?

11] Describe the end behavior when the leading coefficient of an even degree polynomial is positive.	12] Describe the end behavior when the leading coefficient of an even degree polynomial is negative.
13] Describe the end behavior when the leading coefficient of an odd degree polynomial is positive.	14] Describe the end behavior when the leading coefficient of an odd degree polynomial is negative.

Using the patterns and conjectures you wrote above...can you write a possible equation for a polynomial function given the descriptions below.

Equation	Graph	$x = -1$	$x = 0$	$x = 1$	Left End Behavior	Right End Behavior	Sign of Leading Coefficient	Degree
		wiggle 5	bounce 4	none n/a				
		cross	bounce 6	cross				
		4	7	1	down	down		

Exit Ticket

Roots vs. X-Intercepts vs. Zeros	Fundamental Theorem of Algebra	Multiplicity	Behavior at an X-Intercept
• A **root** is found when we set an equation equal to zero and solve for x values. ○ Ex: $y = x^2 - 3x + 2$ $0 = x^2 - 3x + 2$ $0 = (x-2)(x-1)$ $(x-2) = 0 \ \& \ (x-1) = 0$ $x = 2 \ \& \ x = 1 \rightarrow$ ***ROOTS*** • A **zero** is the term used when looking at a function. What value do we set in for x to	• The Fundamental Theorem of Algebra states that **every polynomial of degree *n* will have exactly *n* roots.** • In other words, a polynomials has as many roots as its highest exponent (some may be imaginary) • If a polynomial of degree *n* appears to have less than *n* roots, count the number of	• When *n* roots of a polynomial are located at the same x-value, the roots are said to have **multiplicity *n*.** • Ex: $y = (x-3)^3$ $0 = (x-3)^3$ $x = 3 \rightarrow Root$ *This function has one root at $x = 3$ with a multiplicity of 3.*	• **Bounce** - A graph "touches" or "bounces off" the x-axis if values before and after the x-intercept are have the same sign. Otherwise, a graph either crosses or wiggles. • **Cross** - A graph "crosses" the x-axis (linearly) if the root has a multiplicity of one (similar to a linear function having a degree of 1) • **Wiggle** - A graph "wiggles" or "slides" through the

make the function become zero. ○ Ex: $f(x) = x^2 - 3x + 2$ $x = 2 \ results\ in\ f(x) = \ \boldsymbol{zero}$ $x = 1 \ results\ in\ f(x) = \ \boldsymbol{zero}$ • An **x-intercept** is the term used when looking at a graph. The x-intercepts are the points where y=0. We use **roots** and **zeros** to find x-intercepts. ○ Ex: $f(x) = x^2 - 3x + 2$ (2,0) & (1,0) are **x-intercepts** of the graph of f(x)	multiplicities of repeated roots. The total number of roots (including the multiplicities will equal to n).		x-axis if the root has a multiplicity greater than one. (ie. There is an inflection point at the root itself).

- **Practice Problems:**
  For each problem below, state what behavior occurs at the x-intercepts:

  - $P_1(x) = (x-2)(x+5)^2$

  - $P_2(x) = 2(x-2)(x+2)(x-3)$

  - $P_3(x) = (x+3)^2(x+1)(x-1)(x-5)$

  - $P_4(x) = -\frac{1}{2}x(x+4)^2(x+1)^4(x-2)^3(x+2)$

  - $P_5(x) = x^4 - 9x^2$